

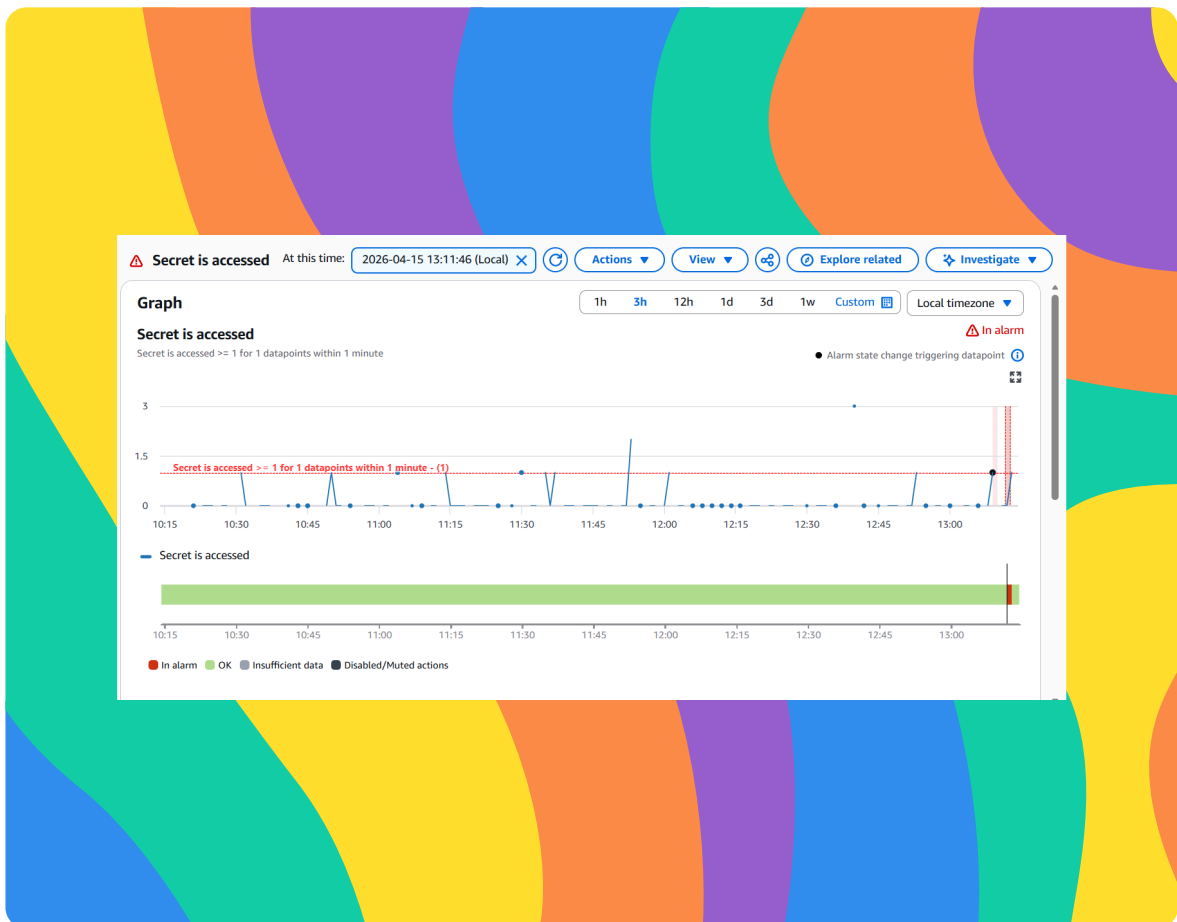


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Build a Security Monitoring System



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Introducing Today's Project!

In this project i'm going to learn how to setup an AWS monitoring system using cloudtrail, cloudwatch and Amazon SNS. This project also demonstrate how to collect logs using cloutrail, processing them using cloudwatch and send alerts using Amazon SNS.

Tools and concepts

Services I used were CloudTrail, CloudWatch, S3 buckets, SNS and Secrets Manager. Key concepts I learnt include securing secrets in the Secrets Manager, monitoring the secret using CloudTrail, analyzing and filtering it using CloudWatch, storing it using S3 buckets and at last get notified using amazon's SNS service.

Project reflection

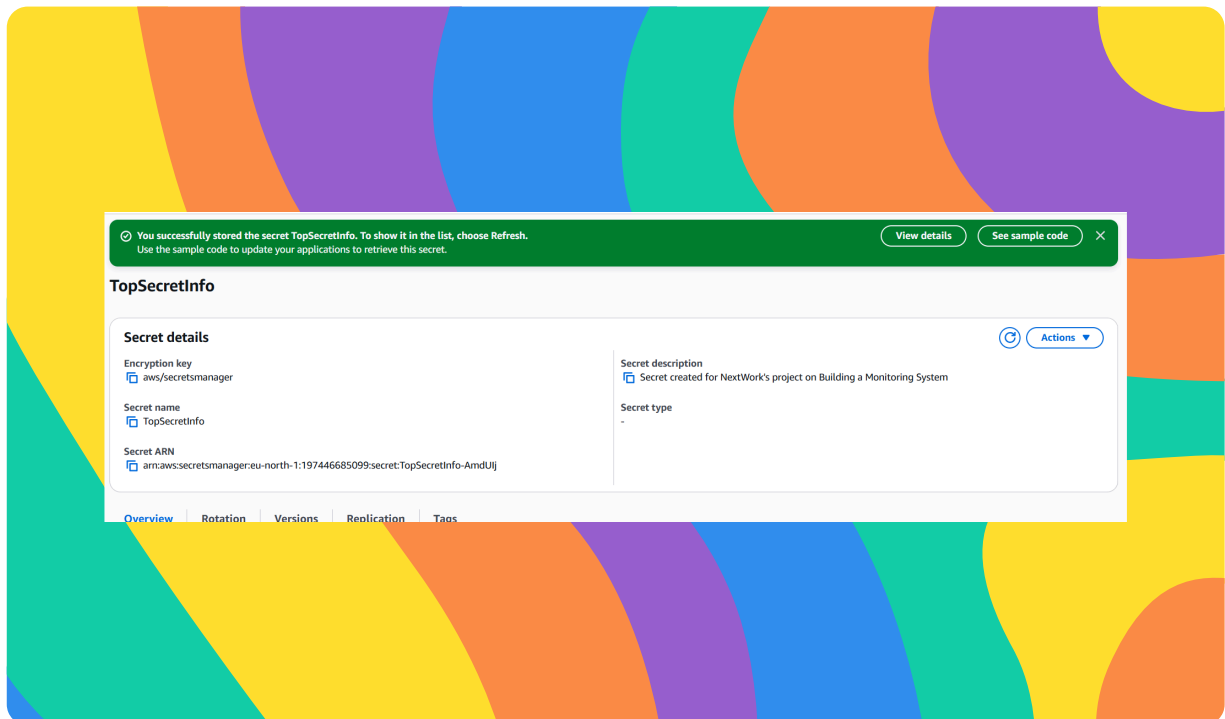
This project took me approximately 3 hours. The most challenging part was the troubleshooting of SNS notification. It was most rewarding too as I learned how to solve the SNS alert problem.



Create a Secret

Secrets Manager is a service from AWS that keeps our sensitive information like passwords, API keys etc. secure and protected. You could use Secrets Manager to store passwords, database credentials, account id's etc.

To set up for my project, I created a secret called "top secretinfo" that contains the value in a key-value pair.





Set Up CloudTrail

CloudTrail is a monitoring service by AWS and it is used for monitoring activities in the AWS account. It records everything like who accessed what, when and where which is very useful for security, troubleshoot and meeting compliance requirements. A trail is like a rule that tell CloudTrail what to record and where to store that record.

CloudTrail events include types like Management events, Data events, Insights events and Network activity events. In this project, we set up our trail to track Management events because accessing secrets fall under Management events. It ain't a Data events because Management events are free to track.

Read vs Write Activity

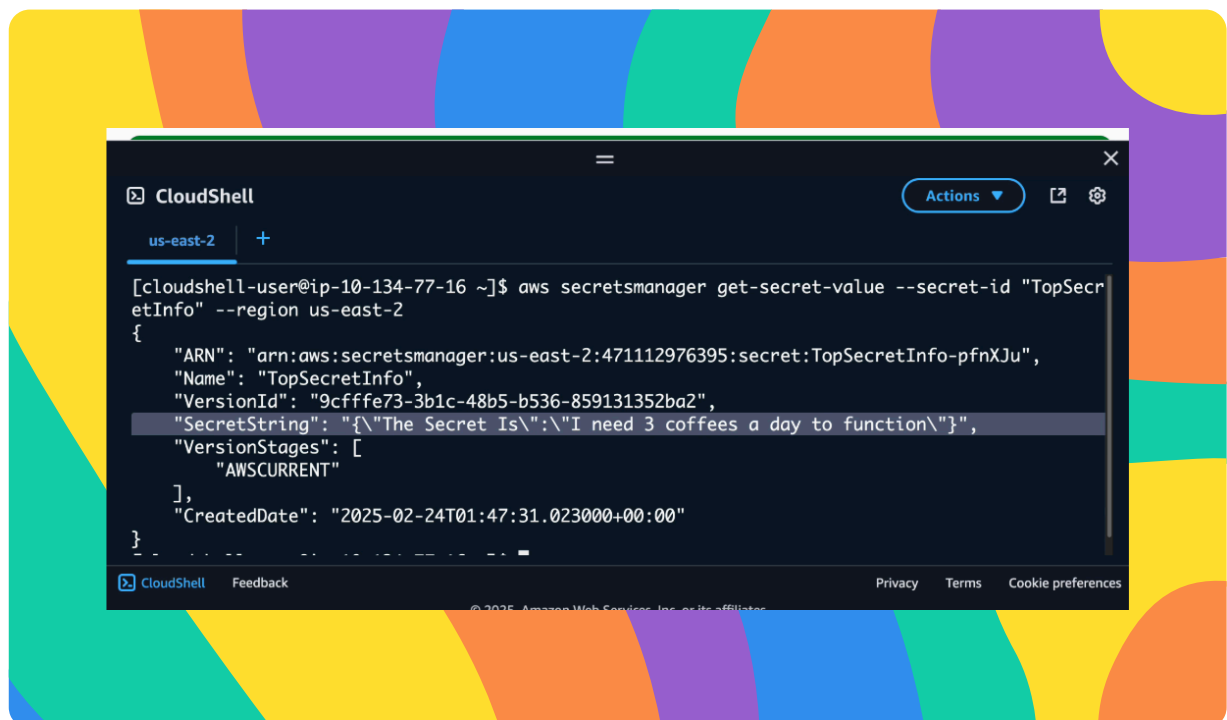
Read API activity involves viewing and but doesn't changing anything like viewing metadata of the secret. Write API activity involves changing like creating, deleting, modifying resources, or even retrieving the value of a secret. For this project, we need to keep read and write API on to track the activities. We could have ticked only Write activity as accessing the secret comes under Write activity.



Verifying CloudTrail

I retrieved the secret in two ways: First through gui console. Second using the terminal

To analyze my CloudTrail events, I visited the event history section to view my logs and when i filtered the logs by lookup attributes. I got the logs which tells that my secret was accessed.



```
CloudShell
us-east-2 +
[cloudshell-user@ip-10-134-77-16 ~]$ aws secretsmanager get-secret-value --secret-id "TopSecretInfo" --region us-east-2
{
  "ARN": "arn:aws:secretsmanager:us-east-2:471112976395:secret:TopSecretInfo-pfnXJu",
  "Name": "TopSecretInfo",
  "VersionId": "9cffe73-3b1c-48b5-b536-859131352ba2",
  "SecretString": "{\"The Secret Is\":\"I need 3 coffees a day to function\"}",
  "VersionStages": [
    "AWSCURRENT"
  ],
  "CreateDate": "2025-02-24T01:47:31.023000+00:00"
}
```

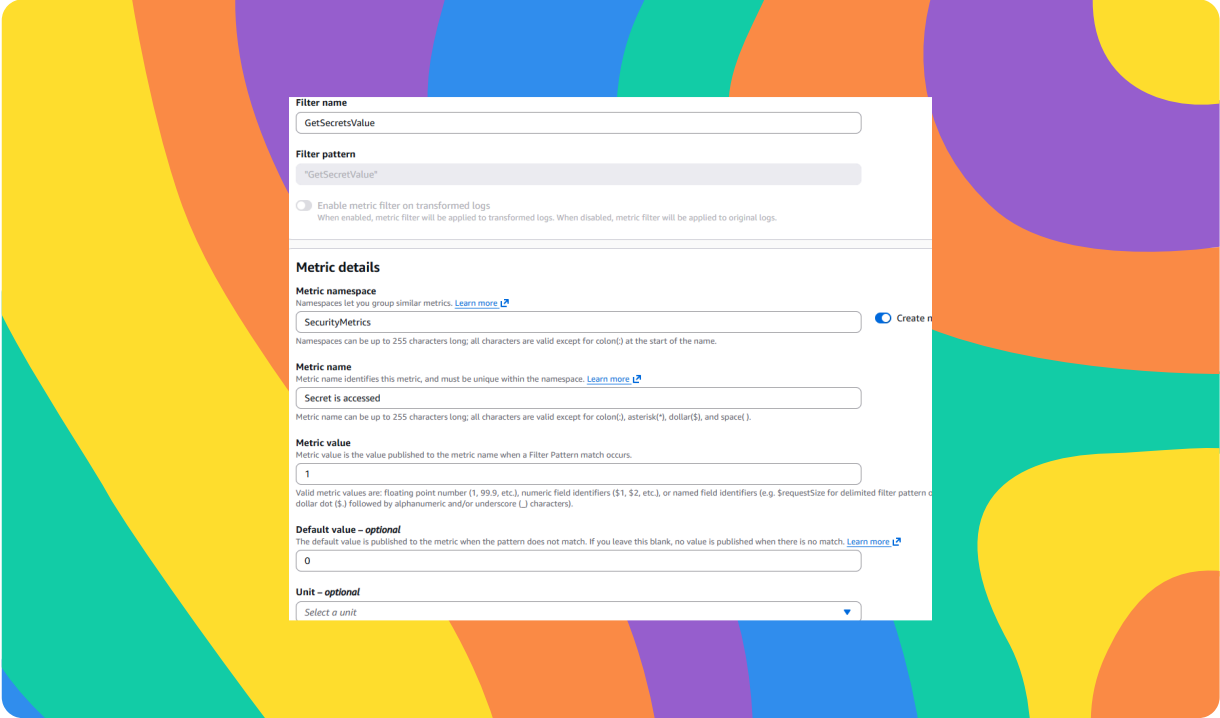


CloudWatch Metrics

CloudWatch Logs is a service from AWS that is used to bring together logs from different AWS services including cloudtrail. It's important for monitoring because these logs can help in analysing, visibility and troubleshooting and when the logs are in cloudwatch we can create desired alerts based on specific patterns. CloudWatch can store different logs such as Application logs, On-premises server logs, Network logs, IoT device logs and CloudFront access logs etc.

CloudTrail's Event History is useful for quick lookup on the recent events and it keeps logs for only 90 days while CloudWatch Logs are better for long term analysing and investigating events. It gives the feature of automated responses when some specific event happen. CloudWatch keeps logs for much time as we want and it as powerful filtering tools.

The purpose of Metric value is when our filter matches in the log i.e if someone accesses the secret the count increases by 1. The purpose of default value is when that when our filter doesn't find any match. Setting this to 0 means that each time no secret access happens it shows zero rather than not showing anything which gives the complete picture means we can see when access happens and when it didn't.



Filter name

GetSecretsValue

Filter pattern

"GetSecretValue"

☐ Enable metric filter on transformed logs

When enabled, metric filter will be applied to transformed logs. When disabled, metric filter will be applied to original logs.

Metric details

Metric namespace

Namespaces let you group similar metrics. [Learn more](#)

SecurityMetrics

Create new namespace

Namespaces can be up to 255 characters long; all characters are valid except for colon(:) at the start of the name.

Metric name

Metric name identifies this metric, and must be unique within the namespace. [Learn more](#)

Secret is accessed

Metric name can be up to 255 characters long; all characters are valid except for colon(:), asterisk(*), dollar(\$), and space().

Metric value

Metric value is the value published to the metric name when a Filter Pattern match occurs.

1

Valid metric values are: floating point number (1, 99.9, etc.), numeric field identifiers (\$1, \$2, etc.), or named field identifiers (e.g. \$requestSize for delimited filter pattern or \$requestSize.\$1 for delimited filter pattern with field identifier). Field identifiers must be preceded by a dollar dot (\$) followed by alphanumeric and/or underscore (_) characters.

Default value – optional

The default value is published to the metric when the pattern does not match. If you leave this blank, no value is published when there is no match. [Learn more](#)

0

Unit – optional

Select a unit



CloudWatch Alarm

A CloudWatch alarm is an alarm which get triggered when the threshold met. I set my CloudWatch alarm threshold to greater or equal to 1 so the alarm will trigger whenever someone accesses my secret even for only one time. The threshold feature is great when it comes to get alerted at any number when secret is being accessed.

I created an SNS topic along the way. An SNS topic is like a broadcasting channel for notifications. First we've to create a channel then we've invite subscribers and then we've to send messages to the topic. My SNS topic is set up to send the notifications on my email address.

AWS requires email confirmation because it wants to confirm the it is the real user, the user the account really belong to. It confirms it through sending an email on the mail inbox of the user, which the user can confirm in the mailbox.



The screenshot displays the AWS Alarms console interface. At the top, there are two notification banners: a green one stating 'Successfully created alarm Secret is accessed.' with a 'View alarm' button, and a blue one stating 'Some subscriptions are pending confirmation' with a 'View SNS Subscriptions' button. Below these, the 'Alarms' tab is selected, showing 'Alarms (1)'. The interface includes a search bar, a 'Quick filters' dropdown, and a table of alarms. The table has columns for Name, State, Actions, Actions muted - new, Last state update (UTC), and Condition. One alarm is listed: 'Secret is accessed' with a state of 'OK' and a last state update of '2026-04-14 12:52:22'. The condition is 'Secret is: 5 minute'.

☐	Name	State	Actions	Actions muted - new	Last state update (UTC)	Condition
☐	Secret is accessed	OK	Loading...	-	2026-04-14 12:52:22	Secret is: 5 minute



Troubleshooting Notification Errors

To test my monitoring system I accessed my secrets manager to check if I get the notification, I found out the i didn't got notified and now i'll troubleshoot this problem and get timely notification.

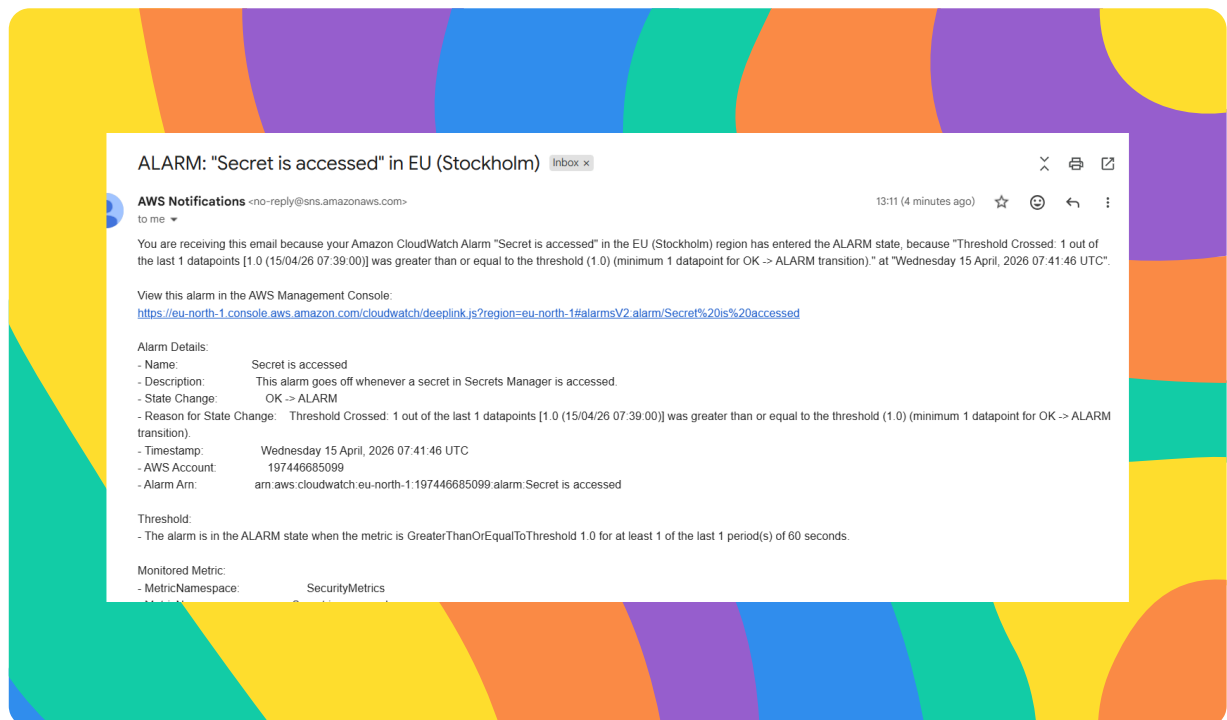
When troubleshooting the notification issues I found found that i was not getting notified when I accesses my secret in the Secrets Manager and after troubleshooting i found out that SNS was not delivering the notification. I corrected it by send myself email via Publish message feature in the SNS. Five ways were:- 1. CloudTrail didn't record the GetSecretValue event. 2. CloudTrail isn't sending logs to CloudWatch. 3. CloudWatch's metric filter isn't filtering logs correctly. 4. CloudWatch's Alarm isn't triggering an action. 5. SNS isn't delivering emails to you.

I initially didn't receive an email before because there was some problem in the SNS service. The key solution was to send email by the Publish message feature in the SNS.



Success!

To validate by accessing the secret. I checked by viewing the alarm. I received email.

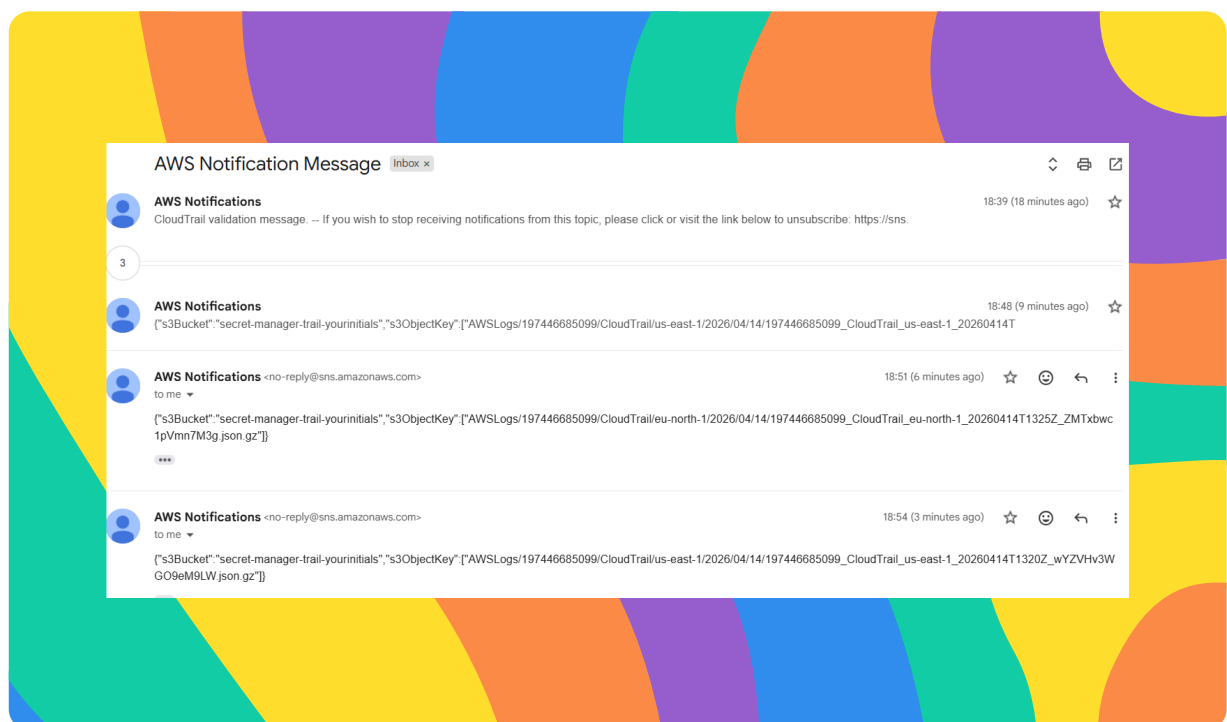




Comparing CloudWatch with CloudTrail Notifications

In the CloudTrail configuration i enabled the SNS notification delivery to recieve notification directly from CloudTrail.

After enabling CloudTrail SNS notifications, my inbox got bombarded with email which is because enabling CloudTrail notification sends alert of every event which might not be useful all the time.





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